THE HOM OBJECT OF A PUSHOUT IS THE FIBRE PRODUCT OF THE HOM OBJECTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the existence and functoriality of the Hom objects is cited from lemma-exists-hom-from-simplicial-set-finite (source lines 1922-1924); that lemma's own proof in the frozen source leaves two checks to the reader (recorded when the same label was reviewed as a card candidate) - the citation here is to the published Stacks lemma and the present proof is complete; the pushout identity is acknowledged 'easy to verify term by term' (source lines 1944-1948), an auxiliary term-level check recorded here.

Editorial note. The label lemma-hom-from-coprod names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Existence and functoriality of the Hom objects. Same chapter, \label{lemma-exists-hom-from-simplicial-set-finite}. Proof lines 1922-1924: all required Hom objects exist and satisfy the correct functorial properties (published Stacks lemma, cited as input).
- Finite limits and coproducts in C. Statement lines 1902-1906. C has coproducts of any two objects and finite limits; U -> V, U -> W morphisms of simplicial sets with finite nonempty terms, degenerate in high degrees.

Source reference. Original reference 017L: sources/stacks-simplicial/pinned-original-source.tex, source lines 1763–1763. Prerequisite pointer invoked at proof line(s) 1923; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Assume the category $\mathcal{C}$ has coproducts of any two objects and finite limits. Let $a : U \to V$, $b : U \to W$ be morphisms of simplicial sets. Assume U_n, V_n, W_n finite nonempty for all $n \geq 0$. Assume that all n-simplices of U, V, W are degenerate for all $n \gg 0$. Let T be a simplicial object of $\mathcal{C}$. Then*

$$\mathrm{Hom}(V, T) \times_{\mathrm{Hom}(U,T)} \mathrm{Hom}(W, T) = \mathrm{Hom}(V \amalg_U W, T)$$

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In other words, the fibre product on the left hand side is represented by the Hom object on the right hand side.

Proof. By Lemma 017L all the required Hom objects exist and satisfy the correct functorial properties. Now we can identify the nth term on the left hand side as the object representing the functor that associates to X the first set of the following sequence of functorial equalities

$$\begin{aligned}
&\mathrm{Mor}(X \times \Delta[n], \mathrm{Hom}(V, T) \times_{\mathrm{Hom}(U,T)} \mathrm{Hom}(W, T)) \\
&= \mathrm{Mor}(X \times \Delta[n], \mathrm{Hom}(V, T)) \times_{\mathrm{Mor}(X \times \Delta[n], \mathrm{Hom}(U,T))} \mathrm{Mor}(X \times \Delta[n], \mathrm{Hom}(W, T)) \\
&= \mathrm{Mor}(X \times \Delta[n] \times V, T) \times_{\mathrm{Mor}(X \times \Delta[n] \times U, T)} \mathrm{Mor}(X \times \Delta[n] \times W, T) \\
&= \mathrm{Mor}(X \times \Delta[n] \times (V \amalg_U W), T))
\end{aligned}$$

Here we have used the fact that

$$(X \times \Delta[n] \times V) \times_{X \times \Delta[n] \times U} (X \times \Delta[n] \times W) = X \times \Delta[n] \times (V \amalg_U W)$$

which is easy to verify term by term. The result of the lemma follows as the last term in the displayed sequence of equalities corresponds to $\mathrm{Hom}(V \amalg_U W, T)_n$. $\square$